Retail Sales & Profitability Analysis

KPI Dashboard Summary and Business Recommendations

Dataset: Sample Superstore workbook (Orders, People, Returns). The analysis is based on the supplied workbook and covers 10,194 order-line records.

1. Executive Summary

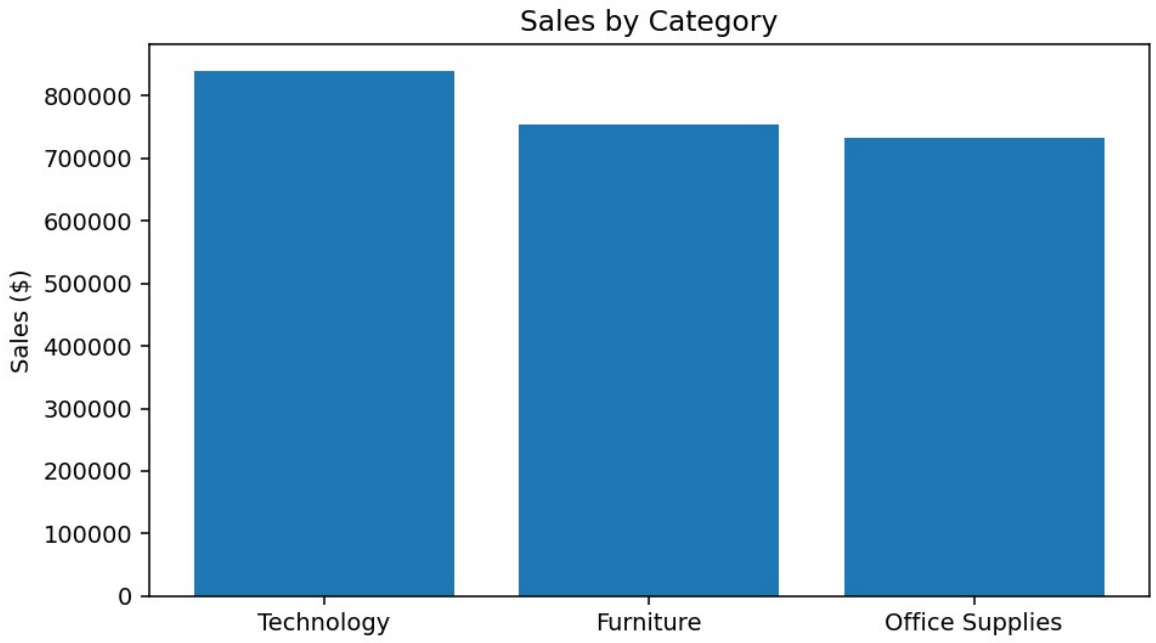
The dataset records $2,326,534.35 in sales and $292,296.81 in profit, giving an overall profit margin of 12.56%. There are 5,111 unique orders and 38,654 units sold. Technology is the leading category by both sales and profit. The West region leads in both sales and profit. Profit is negative in aggregate at several higher discount levels, and Tables is the lowest-profit sub-category.

2. KPI Snapshot

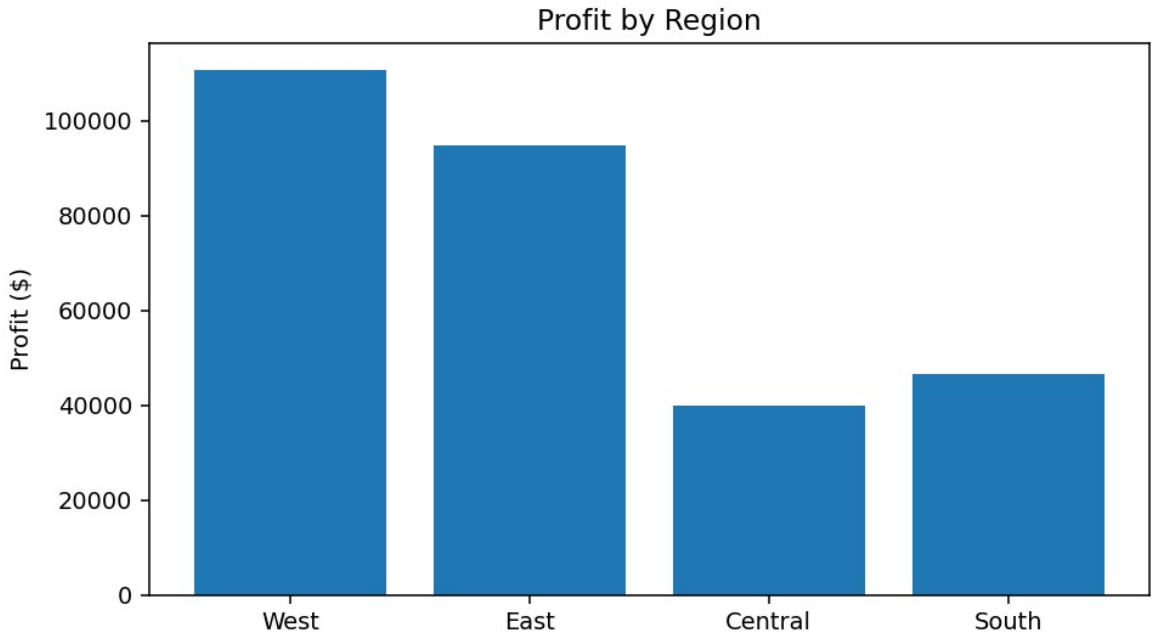
KPI	Result
Total Sales	$2,326,534.35
Total Profit	$292,296.81
Overall Profit Margin	12.56%
Total Quantity	38,654
Best-selling Category	Technology
Most Profitable Category	Technology
Top Region by Sales	West
Top Region by Profit	West
Discount levels with negative aggregate profit	0.3, 0.32, 0.4, 0.45, 0.5, 0.6, 0.7, 0.8
Most loss-making sub-category	Tables
Negative-profit order lines	1,901 (18.6% of order lines)
Unique orders	5,111

3. Category and Regional Performance

Technology generated the highest category sales and profit. Furniture produced the second-highest sales but much lower profit than Technology, so its cost, discounting, and product mix deserve review.

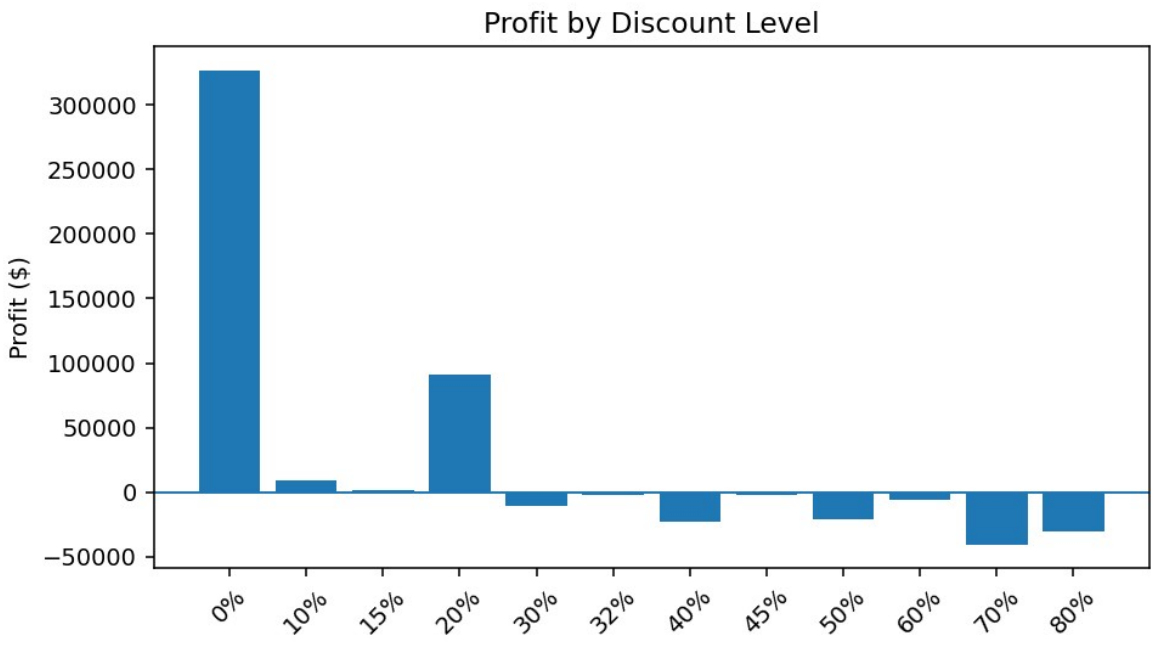


The West region led total sales and profit. The East region was second by sales and profit. Regional comparisons should consider product mix, discount levels, and customer segments before attributing differences to team performance.



4. Discount and Loss-Making Analysis

Discount levels with negative aggregate profit were: 30%, 32%, 40%, 45%, 50%, 60%, 70%, 80%. This is an association in the observed data, not proof that discount alone caused the losses. Review order-level margin, product cost, and discount approval rules at these levels.



There were 1,901 order lines with negative profit (18.6% of all order lines). The Tables sub-category had the lowest aggregate profit ($-17,753.21). Bookcases and Supplies were also among the lowest-profit sub-categories.

5. Returns and Data Quality

The Returns sheet contains 296 unique returned Order IDs. The Orders sheet has 10,194 rows and 10,194 unique Row IDs, with no duplicate Row IDs detected. The key fields checked (Order ID, Order Date, Customer Name, Sales, and Profit) contain no missing values. Postal Code also has 0 missing values in this file.

The order-level return analysis groups order lines before joining to Returns, preventing returned orders from being counted multiple times simply because an order contains several products.

6. Business Recommendations

- Review discount approvals at levels where total profit is negative; use margin thresholds rather than sales volume alone.
- Investigate the Tables sub-category and other low-profit sub-categories for pricing, freight, product cost, and discount patterns.
- Study Technology and West-region practices to identify product mix or customer segments associated with higher profit, then validate whether these practices transfer elsewhere.
- Monitor sales, profit, and profit margin together. A high-sales product or region is not automatically the most profitable.
- Use monthly trends to monitor changes over time and investigate unusual spikes or declines before making operational decisions.

7. Methodology and Limitations

The workbook was checked for missing values in important fields and duplicate Row IDs. Analysis uses grouped totals, unique order counts, a manager lookup from the People sheet, and a join to the Returns sheet. Profit margin is calculated as total profit divided by total sales. Discount findings are descriptive and do not establish causation. The dates shown in this workbook range from 2023-01-03 to 2026-12-30; this report uses the dates as provided and does not assume a different reporting period.

Prepared for internship task submission. Figures are rounded for readability; the accompanying workbook contains detailed tables.